

Data Structures Not Yet Mentioned Above

Additional data-structure-focused options that students may consider, especially when topics or algorithms must not be repeated. The main module should clearly show how the selected data structure stores, organizes, searches, or processes data.

Data Structure	Simple Meaning	Possible System Idea	Main Module
Multiway Tree	A tree where one node can have more than two children	File/Folder Category Manager	Organize folders, categories, or departments
General Tree	A flexible tree with any number of child nodes	School Organization Structure System	Display hierarchy of offices, departments, sections
Expression Tree	A tree that stores mathematical expressions	Simple Formula Evaluator	Convert and evaluate formulas like $(A+B)*C$
Suffix Tree	Stores all suffixes of a word/text for fast text search	Document Text Pattern Finder	Find repeated words or phrases quickly
Suffix Array	Array-based version for suffix/text searching	Keyword Search in Notes System	Search patterns inside stored notes
Rope Data Structure	Stores long text efficiently	Simple Text Editor System	Insert/delete text efficiently in long documents
Quad Tree	Divides a 2D space into four parts	Map Area / Seat Zone System	Manage locations, zones, or spatial areas
Octree	Divides 3D space into eight parts	3D Room/Object Organizer	Represent 3D spaces or objects
KD-Tree	Organizes points based on dimensions	Nearest Facility Finder	Find nearest room, building, or service point
Interval Tree	Stores intervals or ranges	Room Schedule Conflict Checker	Detect overlapping schedules
Segment Tree	Handles range queries efficiently	Sales/Attendance Range Summary	Get total, min, max within date ranges
Fenwick Tree / Binary Indexed Tree	Efficient cumulative total structure	Daily Attendance Counter	Compute running totals quickly
Treap	Combination of tree and heap	Randomized Search Record System	Store records with balanced searching
Cartesian Tree	Combines tree structure with priority ordering	Ranking/Score Structure System	Organize values based on order and priority
Ternary Search Tree	Like Trie, but each node has three links	Word Dictionary Search System	Store and search words efficiently
Radix Tree / Patricia Trie	Compressed version of Trie	Fast Prefix Search System	Search names, codes, or keywords with less memory
Merkle Tree	Tree used to verify data integrity	File Integrity Checker	Check if records/files were changed
Bloom Filter	Space-efficient membership checker	Duplicate Email/ID Pre-checker	Check if an item may already exist
Cuckoo Hash Table	Hash table using two possible positions	Fast Lookup System	Store and search records with collision handling
Multimap	One key can have multiple values	Student Subject Enrollment System	One student linked to many subjects
Multiset / Bag	Allows repeated values and counts them	Inventory Item Counter	Count duplicate items or repeated entries